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# E-Commerce Sales & Customer Behavior Analysis

## Exploratory Data Analysis Report

A data-driven examination of online retail transactions, revenue distribution, product performance, customer purchasing patterns, and cancellation activity.

|         |                                                          |
|---------|----------------------------------------------------------|
| Dataset | UCI Online Retail                                        |
| Period  | December 2010 – December 2011                            |
| Tools   | Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab |
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**Purpose.** This report summarizes the results visible in the completed notebook. Findings are descriptive and depend on the filters and calculations used in that analysis.

DATA ANALYSIS / BUSINESS INSIGHTS / VISUAL EXPLORATION

## 1. Executive Summary

This project explores transaction-level records from the UCI Online Retail dataset to understand revenue patterns, product performance, geographic concentration, customer activity, and cancellations. The notebook reports 524,878 sales lines, 19,960 unique invoices, and 4,338 customers with identified IDs in the sales analysis. Reported gross sales revenue is £10.64 million, with an average order value of £533.17.

|                      |                 |                      |
|----------------------|-----------------|----------------------|
| <b>524,878</b>       | <b>19,960</b>   | <b>4,338</b>         |
| Sales lines analyzed | Unique invoices | Identified customers |
| <b>£10.64M</b>       | <b>5.57M</b>    | <b>£533.17</b>       |
| Gross sales revenue  | Units sold      | Average order value  |

### Headline observations

- **Geographic concentration:** the United Kingdom accounts for 84.59% of recorded gross revenue.
- **Peak month:** November 2011 is the highest-revenue month shown, at £1,503,866.78, with 2,769 unique invoices.
- **Product mix:** DOTCOM POSTAGE is the top description by revenue, so the ranking includes non-merchandise entries.
- **Cancellations and returns:** 1.72% of cleaned lines are flagged as cancelled; 10,587 lines have negative quantities.
- **Revenue relationship:** quantity and line revenue have a Pearson correlation of 0.907.

**Interpretation caution:** reported totals reflect the notebook's filters. Gross sales, cancelled invoices, negative-quantity adjustments, and net revenue should be treated as distinct measures.

## 2. Dataset and Methodology

The analysis uses the UCI Online Retail dataset, containing transaction records from a UK-based online retailer for December 2010 through December 2011. The data were retrieved through *ucimlrepo*, and the features and targets were combined into a Pandas DataFrame.

| Field                   | Meaning                                                                                      |
|-------------------------|----------------------------------------------------------------------------------------------|
| InvoiceNo               | Invoice identifier; cancellation prefixes flag cancellation-related lines.                   |
| StockCode / Description | Product identifier and product description.                                                  |
| Quantity                | Units recorded on the transaction line; negative values may indicate returns or adjustments. |
| InvoiceDate             | Date and time of the transaction.                                                            |
| UnitPrice               | Price recorded per unit.                                                                     |
| CustomerID / Country    | Customer identifier where available and the customer's country.                              |

### Data preparation

- Trimmed whitespace in text fields and standardized relevant columns.
- Converted invoice dates to datetime and quantity/unit price to numeric types.

- Removed exact duplicates and records missing essential transaction fields.
- Created derived fields for cancellation flags, line revenue, year-month, week, hour, and weekday.
- Used a sales subset for sales-focused summaries while retaining cleaned data for cancellation and negative-quantity analysis.

## Tools

Python, Pandas, NumPy, Matplotlib, Seaborn, ucimlrepo, and Google Colab.

## 3. Sales Performance and Geography

The notebook reports gross sales revenue of £10,642,110.80 across 524,878 analyzed sales lines. Average order value is calculated as the mean of each invoice's summed line revenue in the sales subset.

| Country        | Gross revenue (£) | Revenue share |
|----------------|-------------------|---------------|
| United Kingdom | 9,001,744.09      | 84.59%        |
| Netherlands    | 285,446.34        | 2.68%         |
| EIRE           | 283,140.52        | 2.66%         |
| Germany        | 228,678.40        | 2.15%         |
| France         | 209,625.37        | 1.97%         |

## Finding

Revenue is highly concentrated in the United Kingdom. The next countries in the ranking contribute substantially smaller shares.

## Potential business action

Investigate whether international expansion, localized marketing, and country-specific shipping offers could diversify revenue. Compare market size, shipping costs, margin, and return rates rather than revenue alone.

## 4. Product Performance

| Product description                | Revenue (£) |
|------------------------------------|-------------|
| DOTCOM POSTAGE                     | 206,248.77  |
| REGENCY CAKESTAND 3 TIER           | 174,156.54  |
| PAPER CRAFT, LITTLE BIRDIE         | 168,469.60  |
| WHITE HANGING HEART T-LIGHT HOLDER | 106,236.72  |
| PARTY BUNTING                      | 99,445.23   |

DOTCOM POSTAGE ranks first by recorded revenue, followed by several physical product descriptions. Because the ranking includes postage and other non-merchandise entries, it should not be treated as a pure merchandise leaderboard.

A stronger product-performance assessment would separate merchandise from postage/manual entries and compare units sold, revenue, return frequency, and gross margin where cost data are available.

## 5. Monthly Trends

Monthly aggregations were created for revenue, units, and unique invoices. November 2011 is the highest-revenue month shown, with £1,503,866.78 in recorded revenue and 2,769 unique invoices.

The time-series charts show fluctuations rather than a uniform month-on-month increase. The end-of-period drop should be interpreted cautiously because the dataset ends in December 2011 and may contain only partial-month coverage.

### Potential business action

Review seasonal campaigns, stock availability, and fulfillment capacity around the November peak. Compare the same months across multiple years when more data are available.

## 6. Customer Purchasing Behavior

Customer-level summaries were created for records with a known CustomerID. The notebook reports 4,338 identified customers in the sales analysis and ranks customers by gross revenue. The summary includes revenue, order count, units purchased, last purchase date, and average order value.

Customer purchasing value varies. Customer-level metrics exclude transactions without a known CustomerID, so they should not be interpreted as representing every transaction or every buyer.

### Potential business action

Extend the analysis with Recency, Frequency, and Monetary (RFM) segmentation, cohort retention, and repeat-purchase analysis. High revenue alone does not necessarily imply high profitability.

## 7. Cancellations and Negative Quantities

| Measure                                 | Notebook result |
|-----------------------------------------|-----------------|
| Cancelled lines / cleaned lines         | 1.72%           |
| Cancelled invoices / unique invoices    | 14.81%          |
| Lines with negative quantity            | 10,587          |
| Revenue sum for negative-quantity lines | -£893,979.73    |

The notebook flags cancellations using invoice numbers beginning with “C” and separately counts negative-quantity lines. These measures are related but not interchangeable: an invoice-level cancellation rate is not the same as the percentage of lines cancelled.

Negative-quantity records may represent returns or adjustments. Their revenue sum should not automatically be treated as confirmed refunds or as a definitive net-sales adjustment without validating transaction semantics and accounting rules.

The monthly cancellation-line chart shows higher counts in some later months, including October and November 2011. Further analysis by product, country, and customer could help explain the pattern.

## 8. Distributions, Outliers, and Correlation

Histograms and box plots were used to examine Quantity, UnitPrice, and Revenue. The distributions are right-skewed, with a long tail of high values. Selected visualizations cap axes at the 99th percentile to make central distributions easier to inspect; this does not mean those observations were removed from the underlying analysis.

| Variable pair        | Pearson correlation | Interpretation                     |
|----------------------|---------------------|------------------------------------|
| Quantity – Revenue   | 0.907               | Strong positive association.       |
| UnitPrice – Revenue  | 0.137               | Weak positive linear association.  |
| UnitPrice – Quantity | -0.004              | Essentially no linear association. |

Revenue is calculated as  $\text{Quantity} \times \text{UnitPrice}$ , so its correlation with quantity is partly a consequence of the formula. Pearson correlation describes linear association, not causation. Extreme values and inclusion/exclusion rules can also influence correlations.

## 9. Recommendations

**Diversify geographic exposure.** Assess international markets using revenue, order frequency, shipping costs, and margin.

**Separate merchandise and non-merchandise.** Categorize postage/manual entries before ranking physical products.

**Investigate cancellation patterns.** Break cancellation activity down by month, product, country, and customer.

**Improve customer segmentation.** Use RFM and cohort analysis to identify repeat buyers and retention opportunities.

**Strengthen revenue definitions.** Report gross sales, cancellation activity, returns, and net revenue as distinct metrics.

**Improve seasonality analysis.** Compare equivalent periods and account for partial-month coverage.

## 10. Limitations

- The findings are descriptive and do not establish causation.
- Customer-level analysis excludes transactions without a known CustomerID.
- The dataset covers a limited period, so year-over-year seasonality cannot be assessed from this dataset alone.
- Revenue is computed from  $\text{Quantity} \times \text{UnitPrice}$  and does not include cost, margin, tax, or shipping-cost information.
- Cancellation prefixes and negative quantities do not fully establish the business reason for each transaction.
- Reported KPIs depend on the notebook's filters and treatment of returns and cancellations.

## 11. Reproducibility and Deliverables

The analysis was executed in Google Colab using Python and the UCI repository API. The notebook exports the following CSV summaries:

| File                      | Contents                                                       |
|---------------------------|----------------------------------------------------------------|
| kpi_summary.csv           | Headline key performance indicators.                           |
| monthly_sales_summary.csv | Monthly revenue, units, and invoice counts.                    |
| country_revenue.csv       | Revenue totals by country.                                     |
| customer_summary.csv      | Customer-level revenue and purchasing metrics, when available. |

In Google Colab, the files are saved to `/content/outputs/`. Download or archive this directory before the runtime resets.

## 12. Conclusion

The analysis provides a structured view of online retail performance across sales, products, geography, customer activity, and cancellations. The most prominent findings are revenue concentration in the United Kingdom, a pronounced November 2011 peak, uneven customer contribution, and cancellation/negative-quantity activity that merits closer investigation.

The next step is to strengthen the business definitions behind gross and net revenue, separate merchandise from postage, and add customer-retention and profitability analysis. These extensions would help move the project from descriptive EDA toward more decision-ready retail analytics.

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Dataset: UCI Machine Learning Repository — Online Retail  
<https://archive.ics.uci.edu/dataset/352/online+retail>

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